

Reading Assignment 2, AI for Science, Fall 2026

Due Sat, Sept 26. Discussion is Mon, Sept 28.

What to Read

- **Batatia, Kovács, Simm, Ortner, Csányi**, *MACE: Higher Order Equivariant Message Passing Neural Networks for Fast and Accurate Force Fields*. NeurIPS 35, 11423-11436, 2022. [arXiv:2206.07697](https://arxiv.org/abs/2206.07697)
- **Batatia et al.**, *A foundation model for atomistic materials chemistry (MACE-MP-0)*. J. Chem. Phys. 163, 184110, 2025. [doi:10.1063/5.0297006](https://doi.org/10.1063/5.0297006)
- **Creed et al.**, *Six Open Questions in Machine-Learned Interatomic Potential Foundation Models.*, 2026, [arXiv:2606.07327](https://arxiv.org/abs/2606.07327).

Peter will be using Sunday to pull together materials for discussions, so it's fine to submit late on Saturday.

Advice on how to read:

- Budget about four hours for the set, and skip all supplementary material.
- The NeurIPS paper is short but dense. Try to understand it. Document what you don't understand in your Gradescope submission. The goal is to have you see a complete description of the MACE framework.
- The J. Chem. Phys. paper assumes a lot of chemistry background in Section II – get a high level understanding but if the detail is hard to understand then just skim that part and include some questions about what was confusing in what you upload to Gradescope. Don't worry about all of the details in Section VI. The goal is to have you see an impressive application of MACE.
- The Creed paper will be the focus of our discussion. Focus on really understanding a few of the open problems. The goal is to have you understand where the frontier of research is in MLIPs.

The additional instructions below are the same as for Reading Assignment 1.

What to submit

Submit a PDF on Gradescope that contains the following information.

Q1 (Basic Info). What is your name and netid? What paper(s) are you reporting on?

Q2 (Paper Summary). Summarize each paper you are reporting on. If the paper is a research paper, state the paper's claimed research contributions. If the paper is a review or survey paper, state what area they survey, and state some of the major ideas and goals of the area. If you are reporting on multiple papers, provide a separate summary for each one.

Q3 (Paper Comments). Answer at least two of the questions below about

the paper(s) in the reading assignment. If you are reporting on multiple papers you don't need to give separate answers for each paper.

- What did you find interesting or exciting about the paper(s)?
- What do you think the paper(s) could have improved?
- What ideas for future research did the paper(s) inspire?
- What did you find confusing about the paper(s)?

Q4 (optional). Are there any other thoughts you would like to share?

Rules

Read the papers and write your answers **without an LLM**. LLMs are a good tool for getting a quick understanding of many papers, but really understanding a paper requires you to actually read it. Afterward, if you would like to ask an LLM to explain something you did not understand, you may do this, and you may edit your submission if it changes what you want to say.

Reading assignments are **required in every registration option** — 3 credits graded, 2 credits graded, and 2 credits S/U. Students taking the course for 2 credits S/U are required to submit for all but one of the paper discussion days.

Making the PDF

- Any editor works. Export or print to PDF. A legible photo or scan of handwriting is also fine.
- Label your answers Q1, Q2, Q3, Q4. If Gradescope asks you to assign pages to questions, assign the page containing each answer.